Lecture 1 - Introduction & Python Setup

Management Science

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About this Course

About Me



- **Field:** Optimizing & simulating complex systems
- **Languages of choice:** Julia, Python and Rust
- **Interests:** Modeling, simulations, machine learning
- **Teaching:** OR, algorithms, programming & mathematics
- **More:** tobiasvlcek.com



I really appreciate active participation and interaction!

What is Management Science?

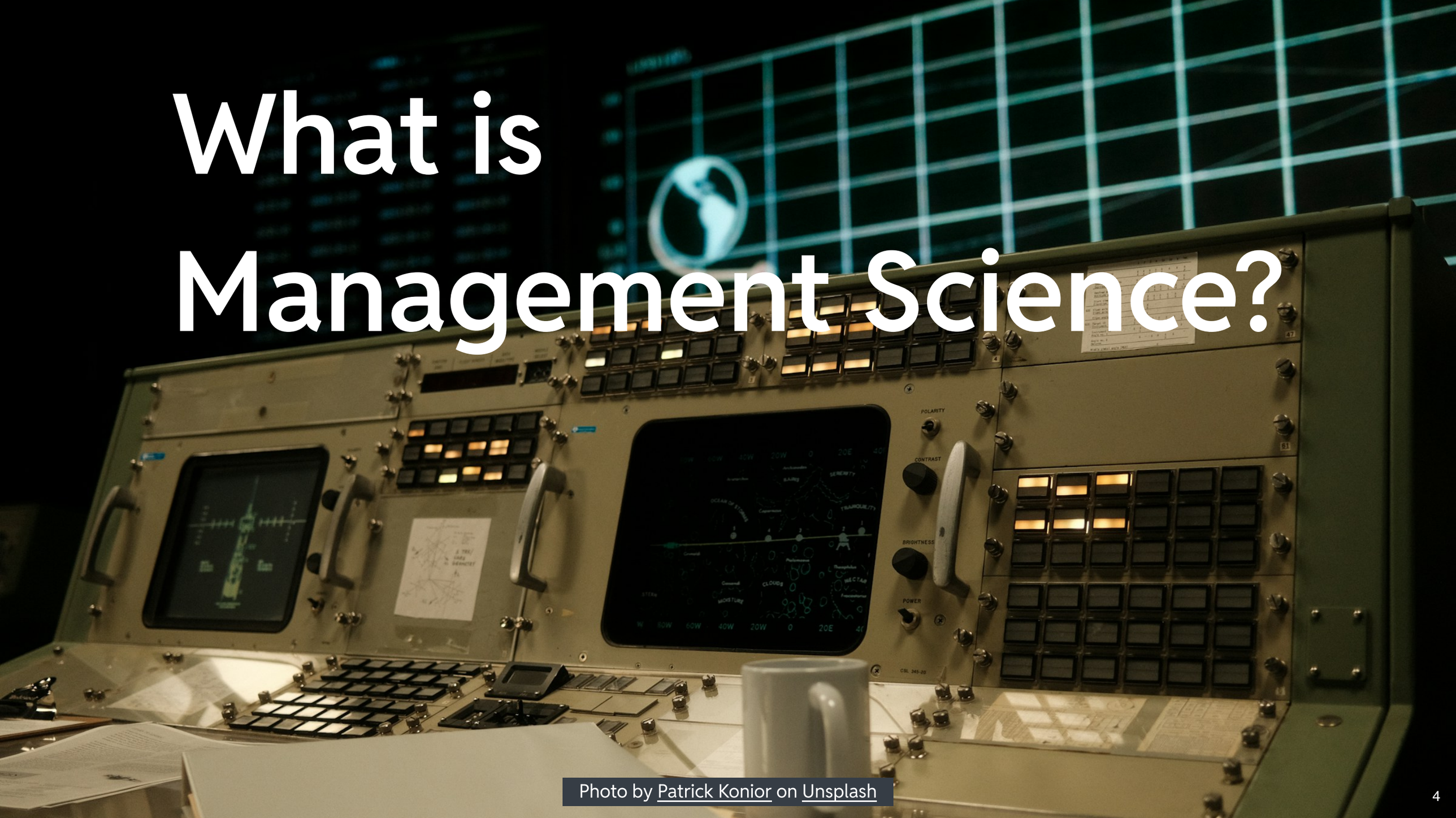


Photo by Patrick Konior on Unsplash

Management Science

Management science is an interdisciplinary field that applies research-based methods, such as modeling, statistics, and algorithms, to solve complex organizational problems and guide strategic decisions, seeking optimal or near-optimal outcomes across institutions, corporations, and governments.[1]

It is not the science of management!

Course Outline

- **Part I:** Python Foundation (Lectures 1-3)
  - Variables, loops, functions, data science tools
- **Part II:** Management Science Tools (Lectures 4-9)
  - Monte Carlo, Scheduling, Routing, Metaheuristics
- **Part III:** Consulting Competition (Lectures 10-12)
  - Client briefings, development, final presentations

> ⚠ **You're now consultants** learning to solve real business problems!

From Amazon to Hospitals

Where Algorithms Make Millions

- **Amazon:** Routing algorithms save millions in delivery costs
- **Uber:** Real-time matching algorithms connect drivers and riders
- **Hospitals:** Scheduling algorithms optimize shifts and OR usage
- **Airlines:** Revenue algorithms price tickets dynamically
- **Manufacturing:** Production scheduling reduces waste and delays

Management Science is everywhere!

This Course Approach

The Consultant Mindset

- You'll work on **business problems**
- Learn to **communicate solutions** to non-technical clients
- Build **practical tools** that solve actual challenges
- Compete in a **consulting competition** with real scenarios

> Think like a consultant: What value does my solution bring to the business?

Course Structure

- 12 lectures of intensive learning
- 3 hours per lecture (with breaks!)
- Interactive format:
  - Concepts & Setup
  - Hands-on Python notebooks
  - Consulting Competition
- Final competition: Apply everything you've learned

Each lecture builds on the previous - attendance is important!

Grading

Grade Composition:

- Course has 100 points, 50 needed to pass
- 2 Assignments: 30 points each
- Final Consulting Project: 40 points

> ⚠ We will have several competitions including our final competition, where you will have the chance to earn bonus points!

Assignments

Assignment Strategy:

- Start with **easy warm-ups** in tutorials
- Progress to **real-world problems**
- All assignments have **clear rubrics**
- Bonus points for **creative solutions**!

> ⚠ **Late submissions:** -10% per day (max 3 days)

Learning Python

- In my experience, the best way to learn is by doing!
- Here, we will focus on decision algorithms
- You will start to learn Python by **doing the tutorials**



Don’t worry, I will help you out if you have any questions!

What to Expect

- No prior programming required - we start from zero!
- **But experience is helpful!**
- Fast-paced but with lots of support
- Practical focus - less theory, more doing
- Mistakes are welcome - they’re how we learn



The course gets easier after week 3 - the hardest part is getting started!

Course Goals

By the end, you will be able to:

1. Write Python code to solve business problems
2. Apply algorithms for scheduling, routing, and optimization
3. Simulate uncertainty using Monte Carlo methods
4. Present solutions like a management consultant
5. Build tools that create business value

You'll leave with a portfolio of ideas to work on real solutions!

Why Python?

- **Origins**: Conceived in late 1980s as a teaching and scripting language
- **Simple Syntax**: Python's syntax is mostly straightforward and very easy to learn
- **Versatility**: Used in web development, data analysis, artificial intelligence, and more
- **Community Support**: A large community of users worldwide and extensive documentation

Help from AI

- You are **allowed to use AI** in the course
- I use it as well (e.g., Claude, ChatGPT, Gemini, …)
- These **tools are great** for learning Python!
- Can help you a lot **to get started** with programming
- I will also teach you how to use it effectively

> ⚠ But you should *not* simply use them to *replace* your learning.

Learning to Program

My Recommendation

1. **Be present**: Attend the lecture and solve the tutorials
2. **Put in work**: Repeat code and try to understand it
3. **Do coding**: Run code, play around, modify, and solve
4. **Compete**: Participate in the competitions to learn



Great resources to start are books and small challenges. You can find a list of recommendations in the literature recommendations.

Don’t give up!

Programming is like learning to ride a bike

- You’ll fall a few times
- It feels impossible at first
- Then suddenly… it clicks!

> ⚠ **Lectures 1-3 are the hardest.** Push through and it gets much easier!

Setting up Python

What is an IDE?

- Integrated Development Environment = application
- It allows you to write, run and debug code scripts
- Other IDEs include for example:
  - PyCharm from JetBrains
  - Zed

Install VS Code

- Download and install from the website
- Built for Windows, Linux and Mac
- Install the Python and Jupyter extension
- **Great! First steps are done**



Unsure on how to work with VS Code and notebooks? Ask me! I'm happy to help you out!

Installation of Python with `uv`

- We will use `uv` to install and manage Python versions
- It works on Windows, Mac and Linux
- It helps us to manage packages and virtual environments
- Now, we all go here and install `uv` and Python



If the installation does not work, **let me know!**

Notebooks with uv

Quick Check

- Have you installed `uv` and initialized the project?
- **Great! Before we continue, check the following:**
  - ☐ You have a folder for the course
  - ☐ You have initialized `uv` with `uv init` inside the folder
  - ☐ You can see a file called `pyproject.toml` in the folder

> ⚠ **Something not working yet? Ask me!**

Using Notebooks

- Now we need to add a kernel to our project
- Run `uv add --dev ipykernel` from your terminal
- Now run `uv add jupyter` in the terminal
- This allows us to use `uv` Python in notebooks
- **Done? Perfect. Now we can start!**

Working with Notebooks

- Now you can download the files from the website
- Just click on one of the sessions and open it
- Select `Jupyter` on the right side
- Download and save the files **to your course folder**
- Open them and select “Open with Jupyter Notebook”



That was the hardest part today!

Any questions so far?

What's Next?

After the break, we'll dive into:

- **Tutorial 1:** Variables and basic data types
- **Tutorial 2:** Lists and loops
- **Tutorial 3:** Conditionals and control flow

> ⓘ **Ready for the tutorials?** Make sure your Jupyter notebook is working before we continue!

The End

That's it for our introduction!

Make sure you have:

- [ ] VS Code installed and running
- [ ] Python environment set up with `uv`
- [ ] Jupyter notebooks working
- [ ] The tutorial files downloaded



Take a **10-minute break**, then we'll start with the interactive notebooks.

Literature

Literature on Algorithms

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[1]
- Ferguson, T.S. (1989) 'Who solved the secretary problem?', Statistical Science, 4(3). doi:10.1214/ss/1177012493.

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Here
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.



Think Python is a great book to start with. It's available online for free. Schrödinger Programmiert Python is a great alternative for German students, as it is a very playful introduction to programming with lots of examples.

More Literature

For more interesting literature, take a look at the literature list of this course.